Working with IntelliJ IDE

Student Workbook 1a – Integrated Development Environments

Version 7.1 Y

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Module 1

IntelliJ Basics

Section 1–1

The IntelliJ IDE

Understanding IntelliJ IDE

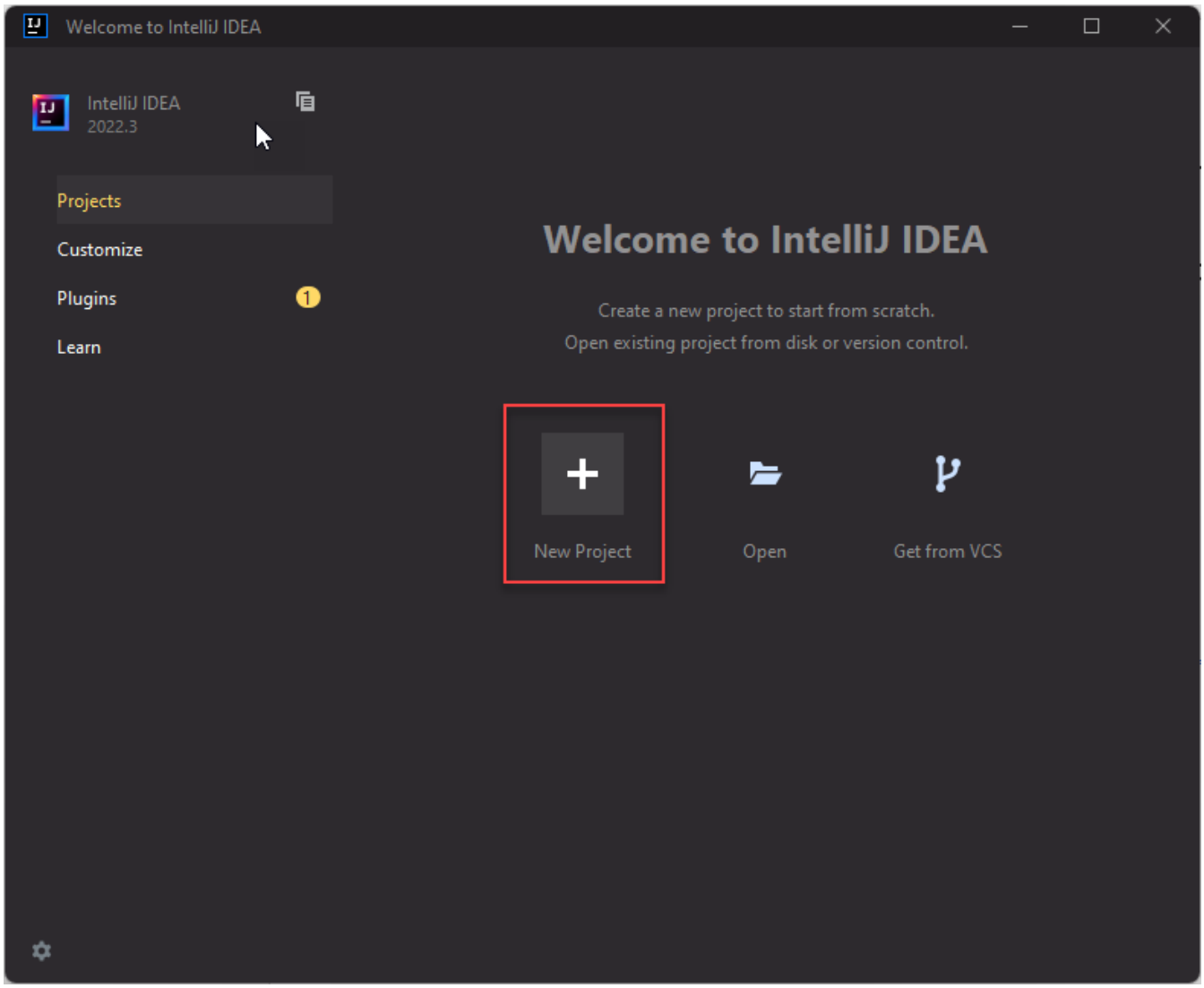
- **Integrated Development Environment - A program that contains comprehensive utilities to develop software**
- **The leading IDE for Java and Kotlin development**
  - Run, Test and Debug Java projects
  - Integrated with Git
- **Plugins available to customize and extend your environment**
- **A product of JetBrains**
  - Free Community Edition or paid Ultimate Edition

Java Projects

- **A Java Project is a folder that contains all project files**
- **Java source code**
- **IntelliJ manages all types of Java projects and build tools**
  - There are a few different project build management tools
    - Maven – currently the most popular build manager for java
    - Gradle
    - Ant
- **Multiple ways to create projects**
  - Create the project directly from IntelliJ
    - IntelliJ will build the appropriate folder structure and starter files
  - Import a project from a VCS such as Git
  - Create a project manually, then open it in IntelliJ
    - Projects can also be created with tools like Maven

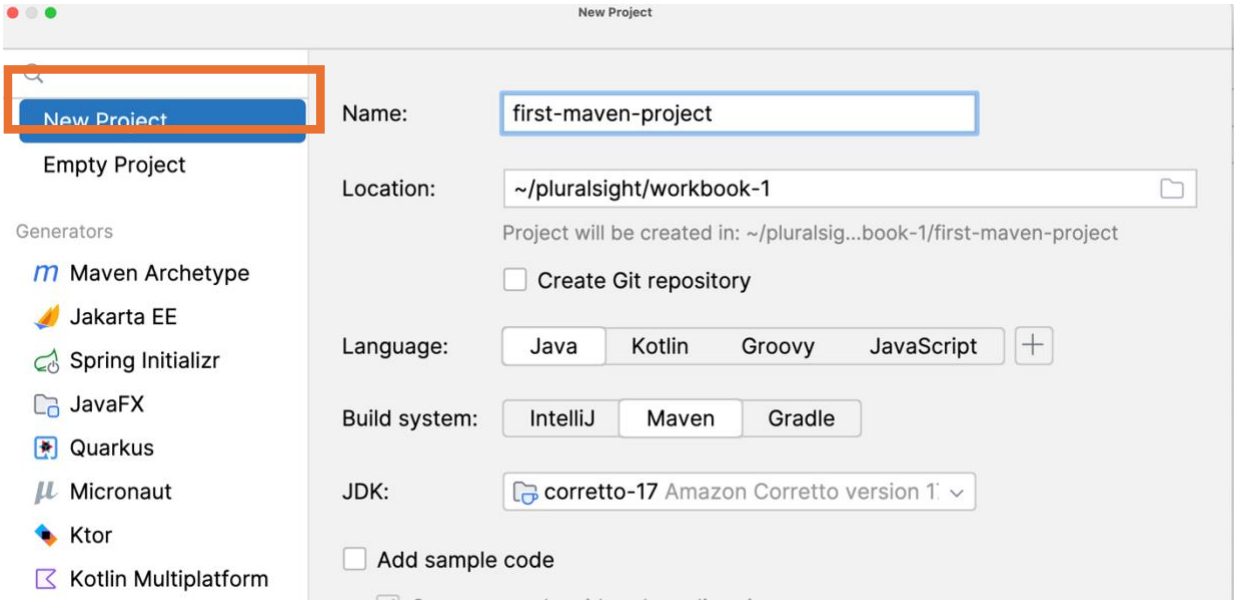
Creating a New Java Project

- **Open IntelliJ and select Projects -> New Project**



The New Project Dialog

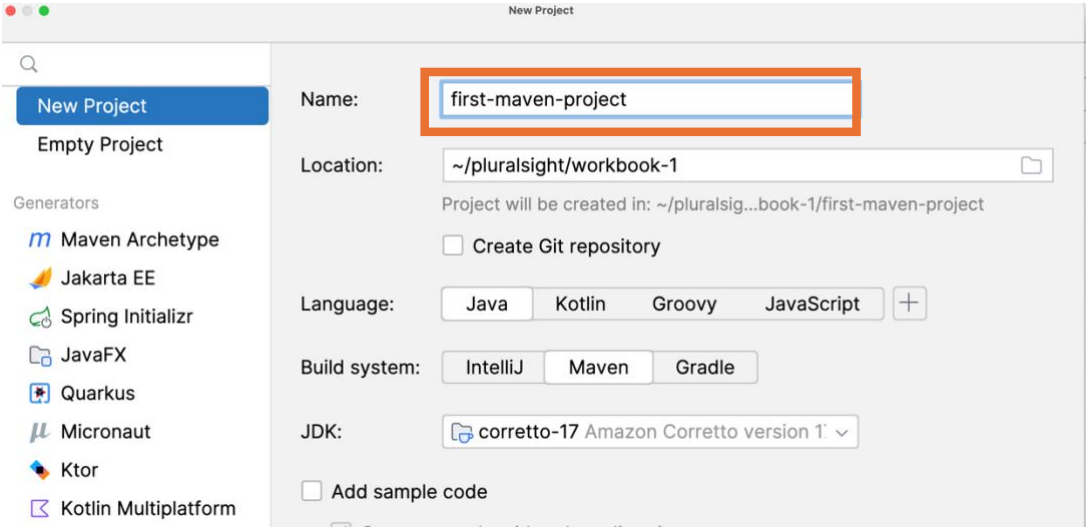
- **The New Project dialog lets you choose from various ways to create and initialize your project**



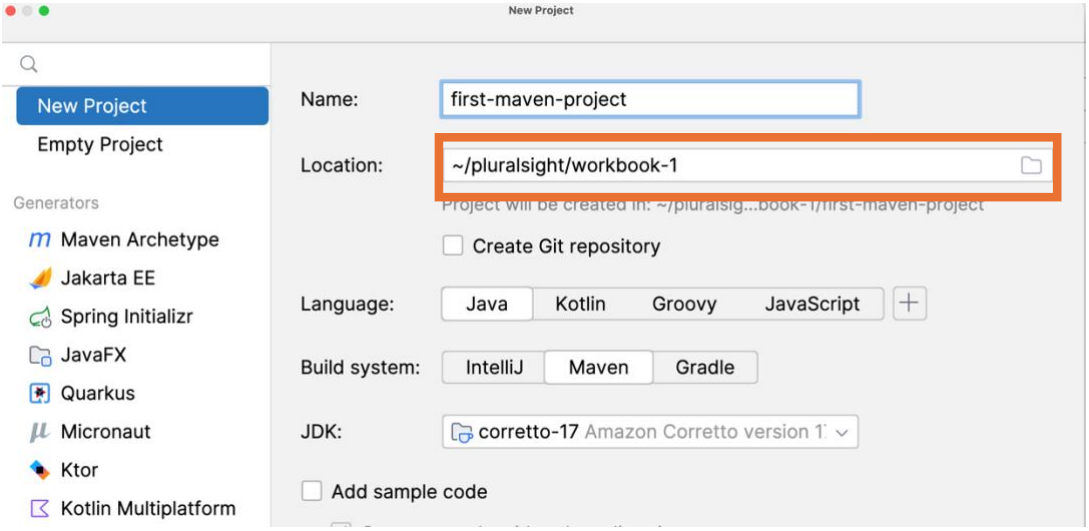
  - New Project creates a basic project with some minimal starter code
  - Empty Project creates the project folder only; all other configurations must be added manually later
  - Generators are specialized project templates
    - These allow you to create projects with significant pre-generated boilerplate (or starter) code

Project Name and Location

- **Enter a Project Name**



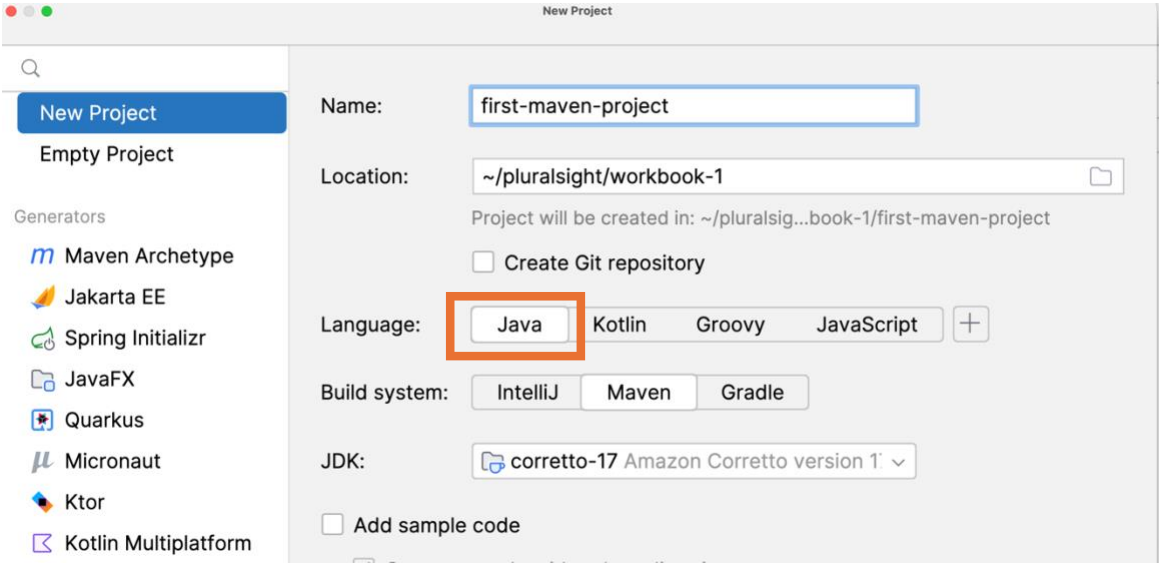
- **Select the directory/folder where the project will be saved**



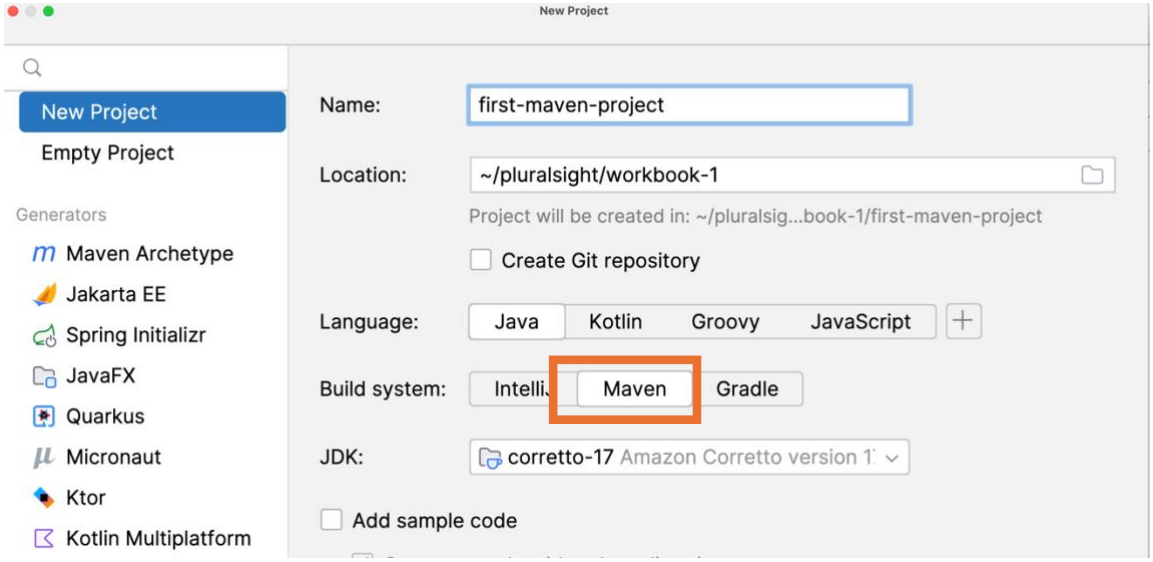
- **You can choose whether to create a local Git repository to hold the project code**

Project Language and Build System

- **Select the project language**

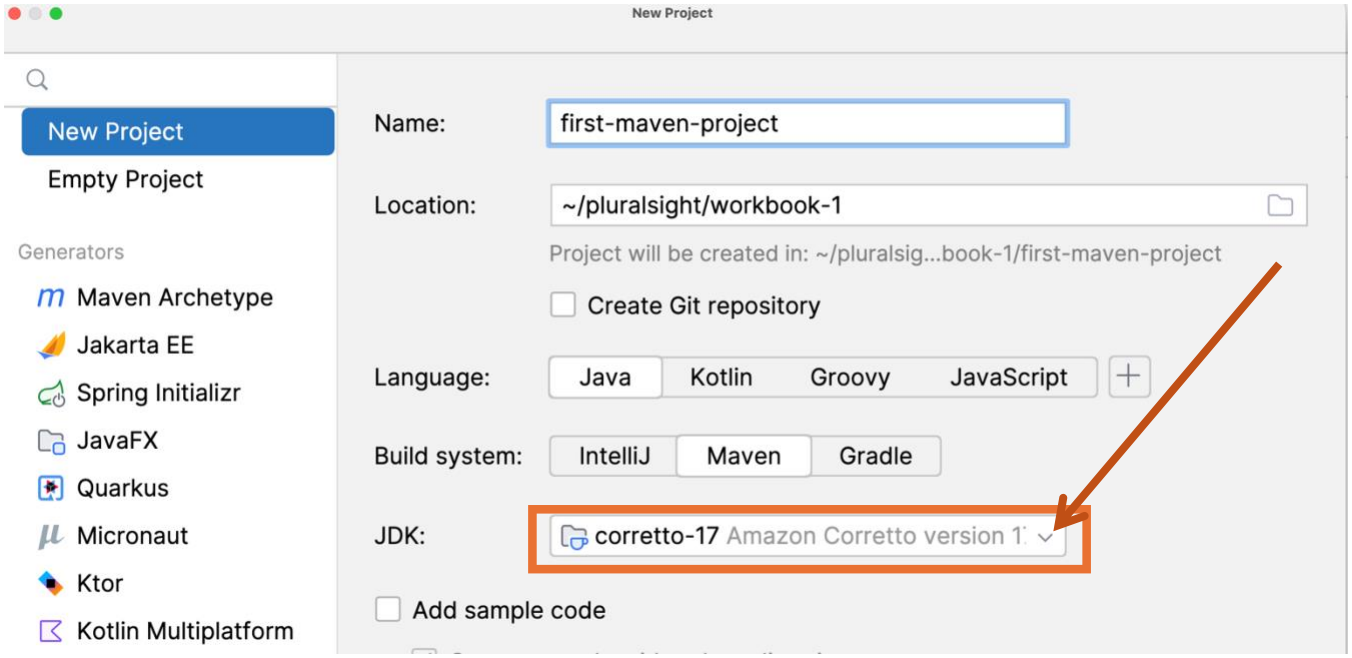


- **Select the project build system**



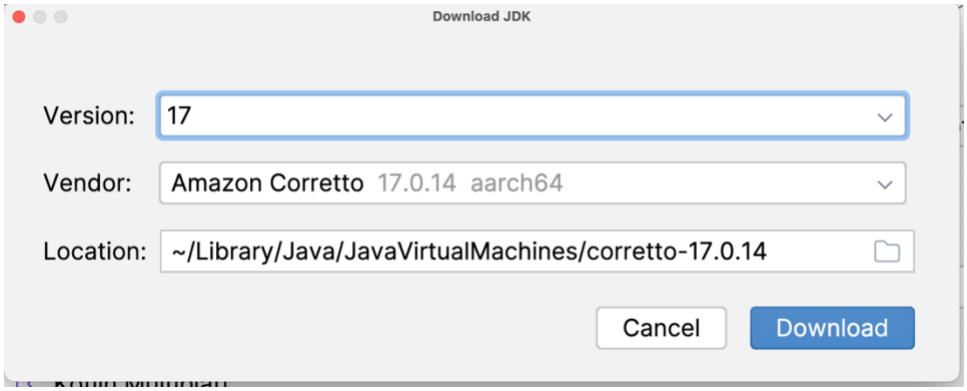
Select the Java Version

- **We will be using Java 17 during this course**
  - If you do not have the Java 17 JDK, IntelliJ will give you options to download and install it directly

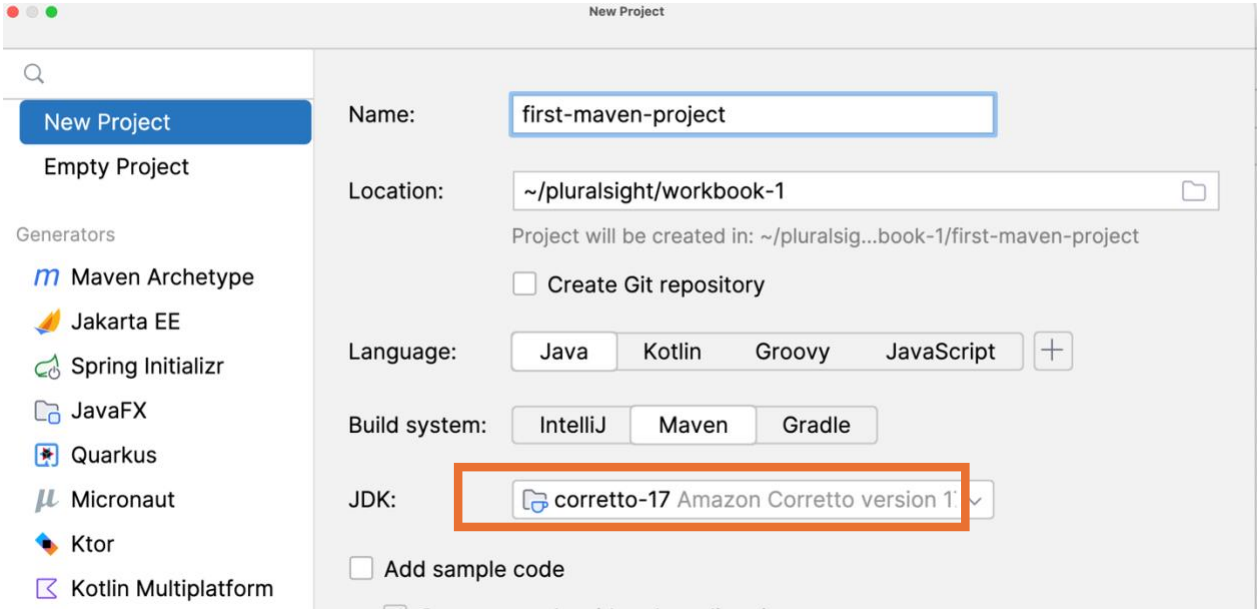


Installing JDK 17 (If Necessary)

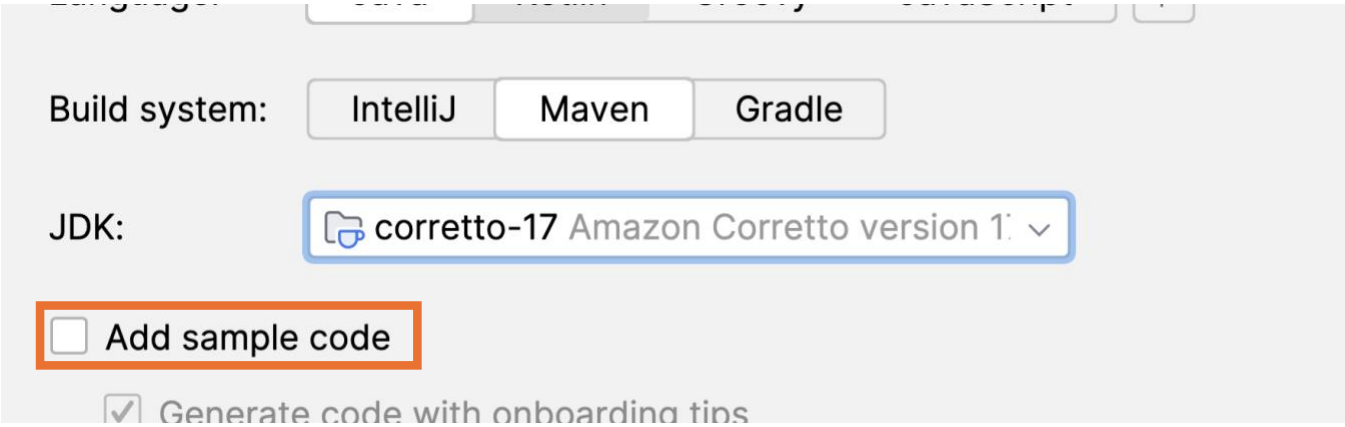
- **Select and download the Java 17 JDK**



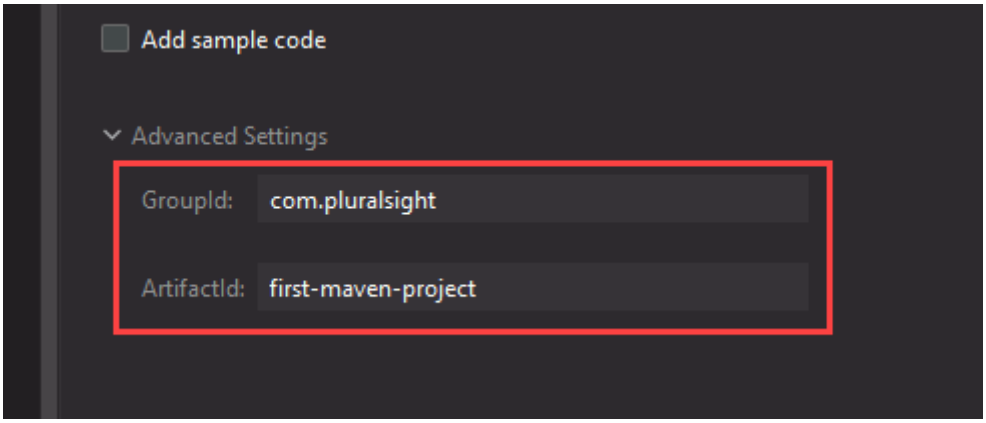
- **Ensure that JDK 17 is selected**



- **Uncheck Add sample code**



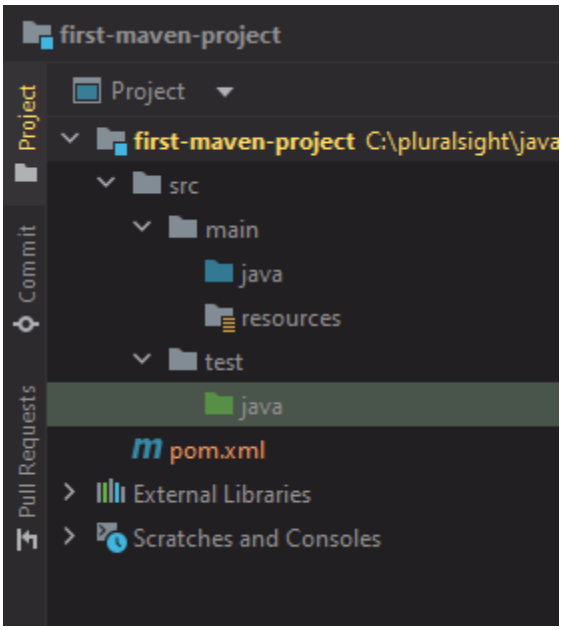
- **Expand the Advanced Settings tab and update the GroupId and ArtifactId**



- **Click Create**

Explore the Project

- **After creating the project IntelliJ will open the project folder**



- **This folder structure is the standard structure for Maven Projects**

```
First-maven-project/
    ├── pom.xml
    └── src/
        └── main/
            └── java/
        └── test/
            └── java/
```

The `pom.xml` File

- **The `pom.xml` file is a Maven file that is used to define**
  - project configurations (name, version, jdk build version, etc.)
  - a list of external project dependencies

pom.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>

    <groupId>com.pluralsight</groupId>
    <artifactId>first-maven-project</artifactId>
    <version>1.0-SNAPSHOT</version>

    <properties>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    </properties>

</project>
```

- **We will learn more about this file later in the cohort**

Maven Project Folder Structure

- **All application code must be added to the** `src/main/java` **directory**

- **Unit tests are added to the** `src/test/java` **directory**
  - You will learn more about unit tests later in the cohort

Adding a Package

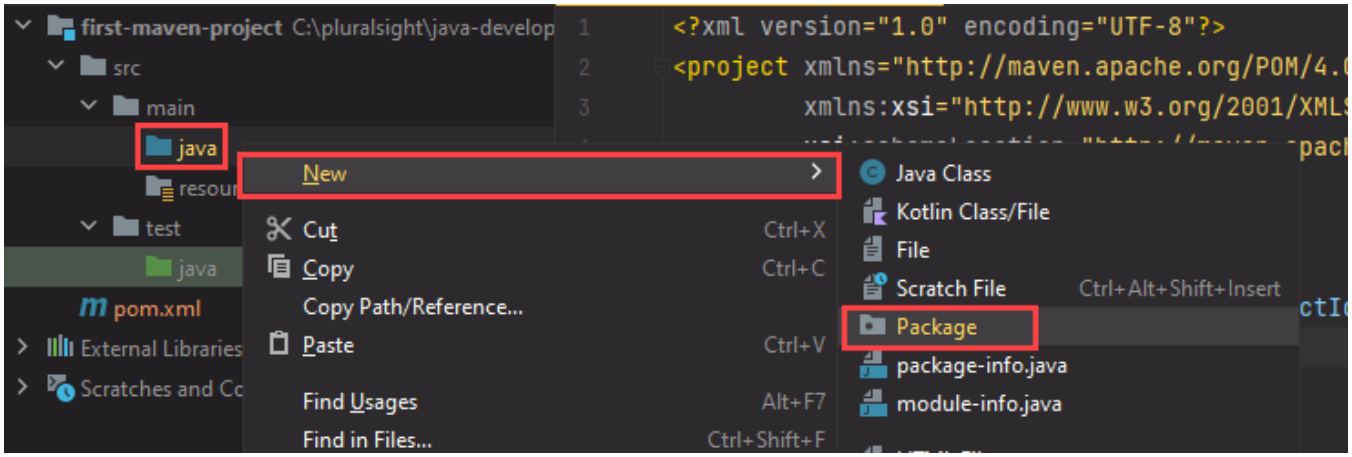
- **Packages appear as folders in a Java project**
  - They allow us to organize our code
  - Packages names are all lower case and follow the following convention

    `com.companyname.projectname`

- **Each dot in the package name implies a subdirectory in the java source tree**

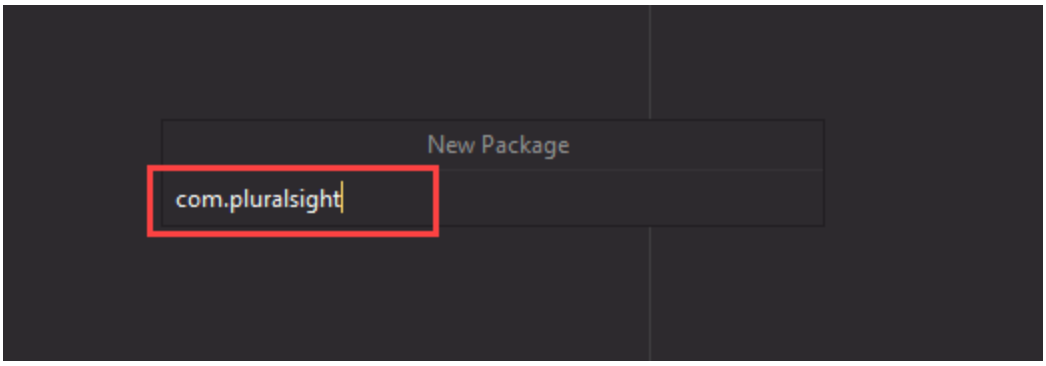
- **Packages are added relative to the `src/main/java` folder**
  - All Java projects *should* have at least one package
    - i.e. we should not add a Java file directly into the `java` folder

- **Create a package by right-clicking on the `main/java` folder and select New -> Package**

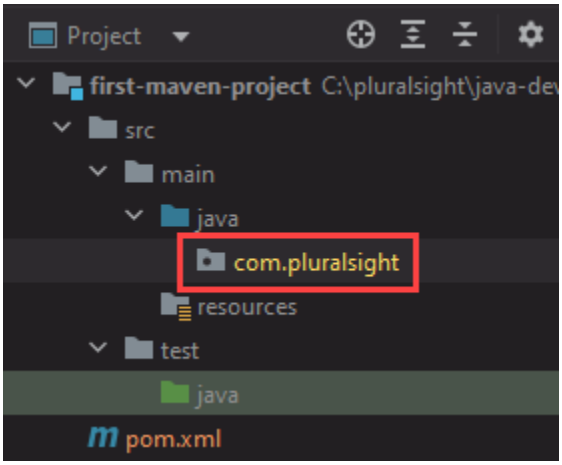


Set the Package Name

- **Add a package name in the New Package window and hit Enter**



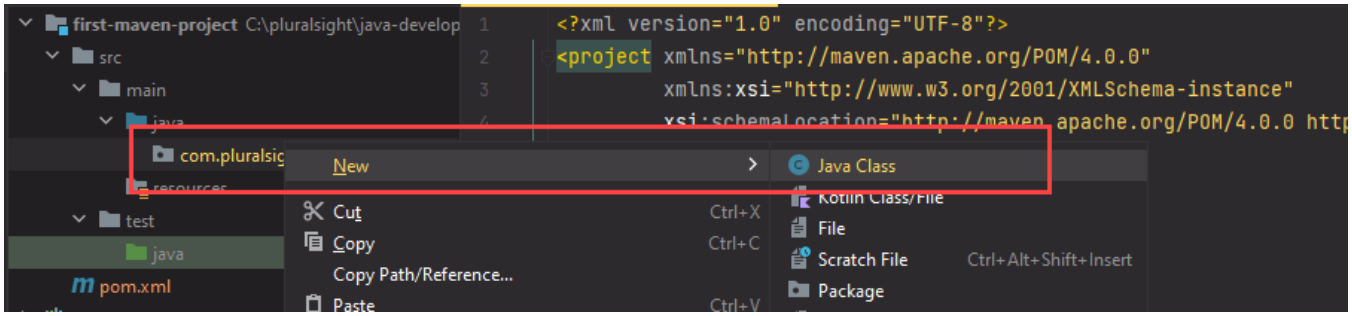
- **A new Package will have been created for you**



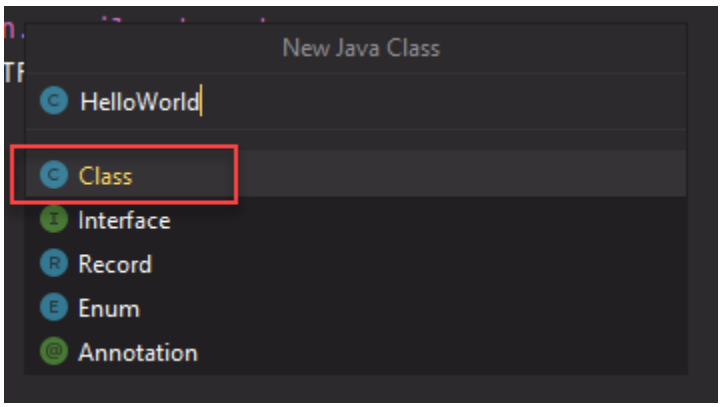
Creating a Class

- **Now that you have created the package, you can create a class in the package.**

- **Right-click on the package you just create in the Project Explorer on the left hand window. Select New -> Java Class**

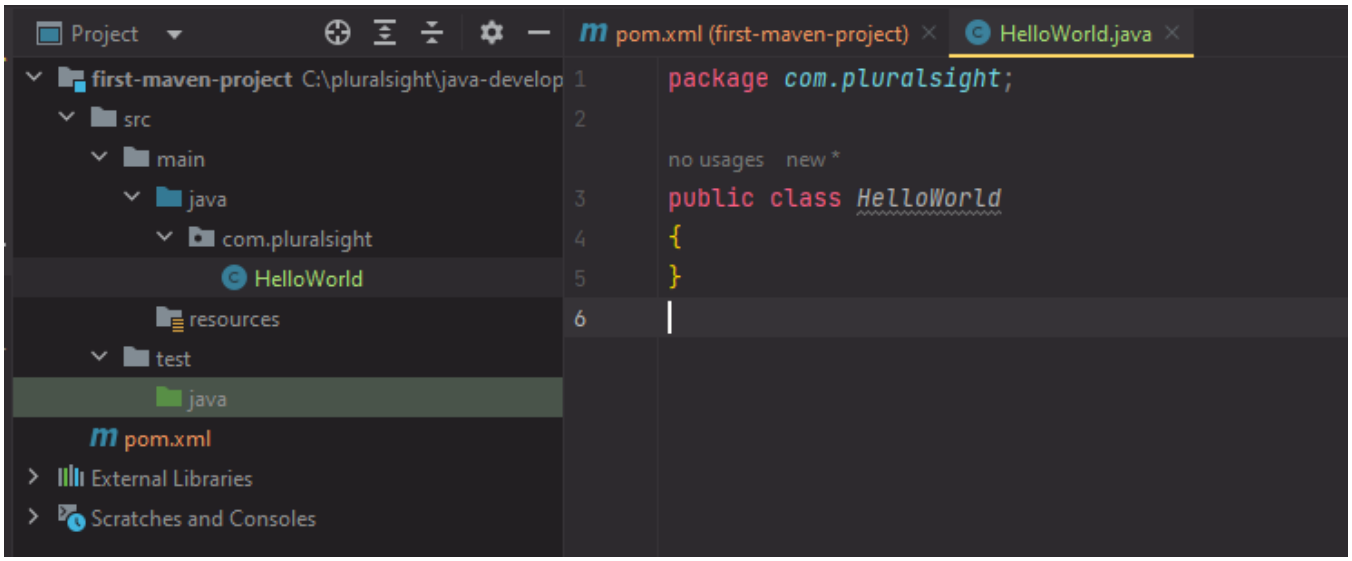


- **In the New Java Class window enter the name of your class and hit Enter**
  - Class names always start with an uppercase letter



The Java Source File

- **Your new class will be created and open in IntelliJ**



Finishing the Application

- **A Java application must have an Entry Point for the application to run it**

- **The Entry Point is a function named `main`, defined like this**

```
public static void main(String[] args) {
   // your code goes here
}
```

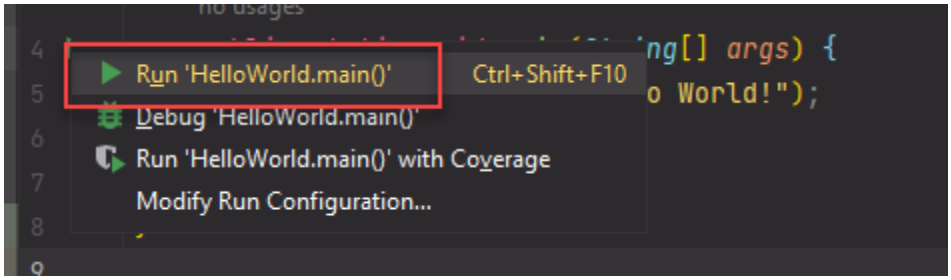
- **Finish the HelloWorld project**

```
HelloWorld.java
1   package com.mycompany;
2
    no usages
3   public class HelloWorld {
        no usages
4       public static void main(String[] args) {
5           System.out.println("Hello World!");
6       }
7
8   }
9
```

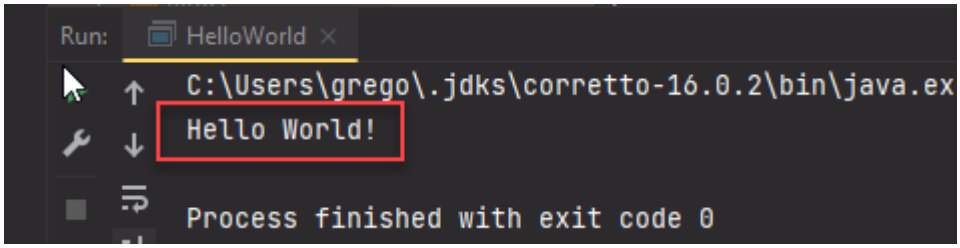
Running Your Application

- **Once you have added the `static void main` function, a green arrow appears next to that function**

- **At any time, you can click on the green arrow, to run your application**

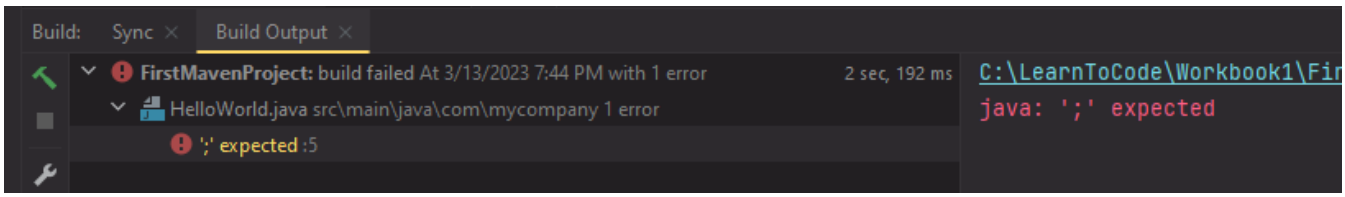


- **This should immediately print the results at the bottom of the window**



Detecting and Fixing Errors

- **If your application does not run, it is most likely because of an error in your code**
  - IntelliJ can help find errors quickly
- **When you attempt to run the application, you may get a compile error message**



Common Errors

- **Misspelled functions or variables**
  - Missing “r” in `println()`

```
no usages
public static void main(String[] args) {
    System.out.pintln("Hello World!");
}
```

- **Missing semi-colon at the end of a line**

```
no usages
public static void main(String[] args) {
    System.out.println("Hello World!")
}
```

- **Missing close curly brace**
  - Here the compiler believes that the function has a close curly, but the class is missing one

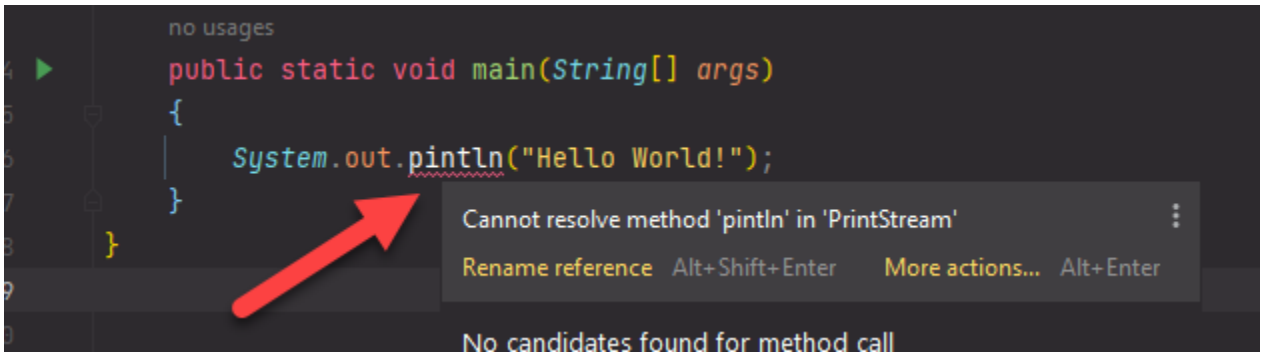
```
public class HelloWorld {
    no usages
    public static void main(String[] args) {
        System.out.println("Hello World!");

}
```

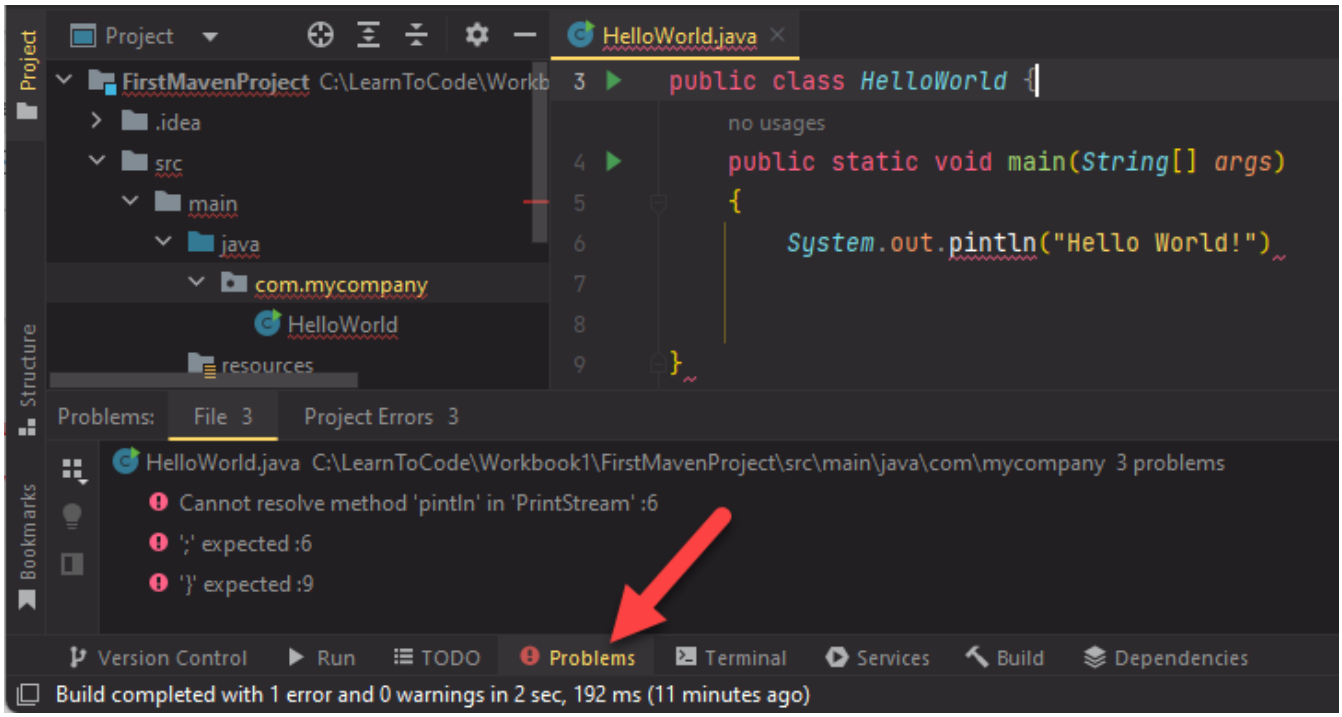
Lean on Your Tools

- **DO read the error messages; they really want to help!**

- **You can also hover over the red squiggly line to get information about the error**



- **Additionally, there is a Problems tab at the bottom of IntelliJ that will list all potential problems with your code**

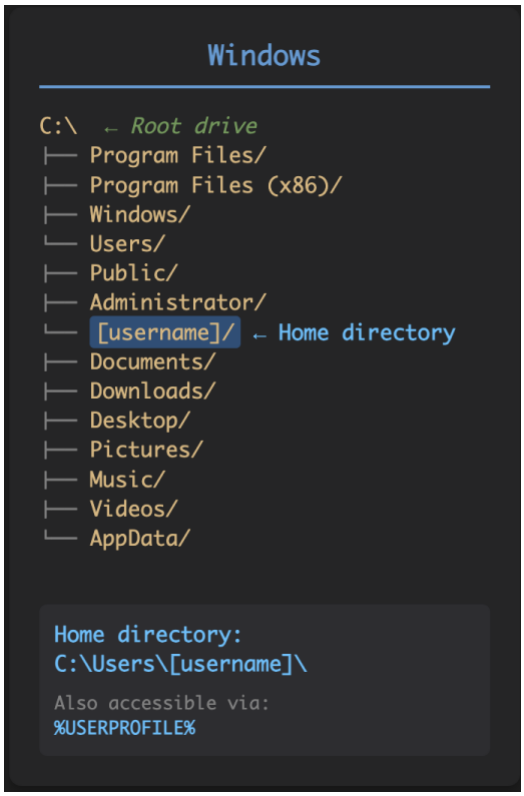


Exercise – Shopping List

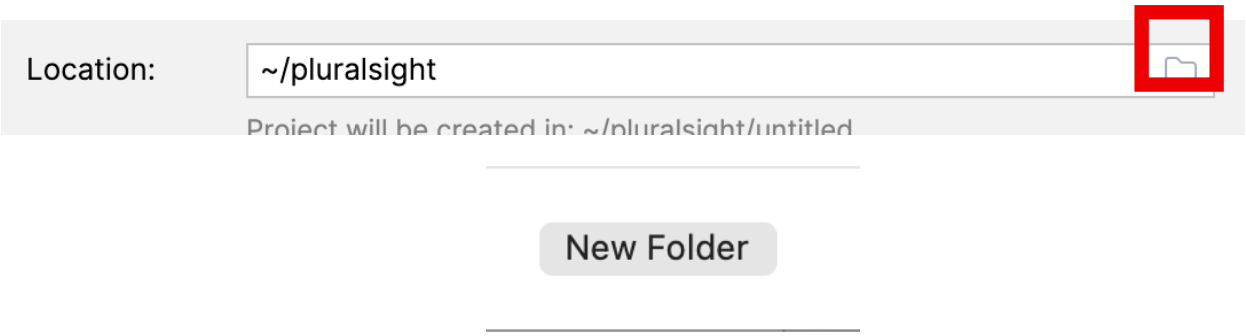
Complete the following exercise by adding the new project into the `workbook-1` folder

If you don't have a `workbook-1` folder, create a new folder within the `pluralsight` directory. The path should look like:

On Mac/Linux: `~/pluralsight/workbook-1`
On Windows: `C:/Users/[username]/pluralsight/workbook-1`



Also, within Intellij's new project screen, the Location folder button will allow for you to navigate to your home directory and create a new folder!



EXERCISE 1

Using IntelliJ, create a new Java application that will list at least 10 items that should be on your shopping list.

1. Name the project `shopping-cart`
2. Create a new package named `com.pluralsight`
3. In the `com.pluralsight` package create a new java class named `ShoppingList`. Remember it must be in a .java file of the same name.
4. Within the `ShoppingList` class, create a `main` method.
5. In the `main()` method use the `System.out.println()` to display a shopping list with at least 10 items. EX. `System.out.println("apricot");`
6. Run the program. If there are any errors, fix them and run it again.
7. Commit and Push your changes to GitHub using your IntelliJ IDE